Electronics guide: MC-6 centrifuge and MW-1 microwave

How to wire both mock appliances to an **Arduino Uno with a Seeed Grove Base Shield v2**, one board per appliance. Each board reads its appliance's buttons, sensors and dial. It drives the display (and, on the centrifuge, the light ring), and reports everything over USB serial to the laptop that runs the verifier. This replaces the Raspberry Pi Pico wiring in the earlier READMEs and manuals.

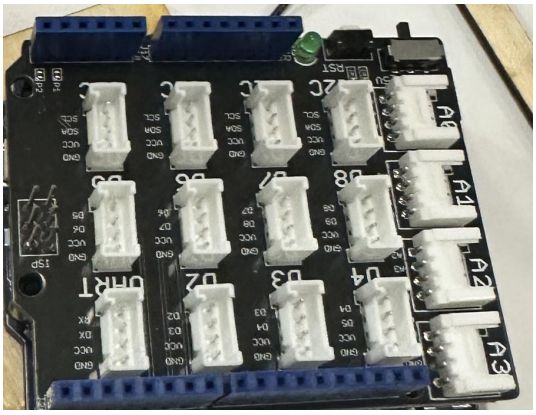
What's in this guide

- **Section 1:** the board, its Grove ports, and the rules that keep wiring simple.
- **Section 2:** one card per part type: what it is, which pins it has, how to connect it, and how to check it.
- **Section 3:** MC-6 centrifuge: port map, wiring table, step-by-step order, schematic, test sketch.
- **Section 4:** MW-1 microwave: the same.
- **Section 5:** testing, troubleshooting, power, and running both from one board.

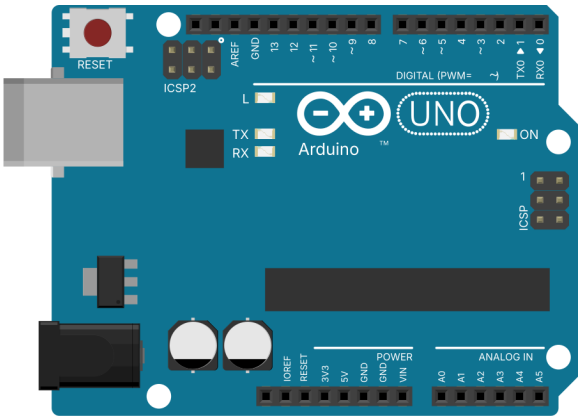
Tools and supplies

Item	Qty	Used for
Soldering iron + solder	1	button lugs, NeoPixel pads
Stranded hookup wire, 22–26 AWG (red, black, one colour per signal)	~3 m	panel buttons, ring
Heat-shrink tube or electrical tape	—	covering every soldered joint
Grove-to-male jumper cables (4-wire)	4 (MC-6), 3 (MW-1)	from shield ports to the breadboard (MC-6) or the button wires (MW-1)
Male-to-female jumper wires	4	OLED header to the Uno header (centrifuge)
330 Ω resistor (orange · orange · brown), optional	1	NeoPixel data line (centrifuge)
Grove Light Sensor v1.2 + Grove cable	2 + 1	lid and tube (centrifuge), door (microwave)
Grove Buzzer + Grove cable	1	microwave
16 mm panel pushbuttons	3 + 5	centrifuge START/STOP/TIME+, microwave all five
Half-size breadboard	1	MC-6 only: 5 V / GND rails, sits outside beside the Uno
USB-B cable	1 per appliance	power and serial to the laptop

1 The board



Your Grove Base Shield v2, plugged onto the Arduino. Photo: your part.



The Arduino Uno underneath, with its header labels. Drawing: Fritzing (CC-BY-SA 3.0).

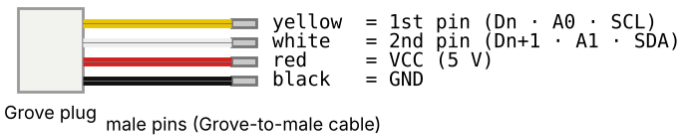
The shield sits on top of the Uno and passes every Uno pin straight up to its blue sockets. It adds white 4-pin Grove ports so modules plug in with one cable instead of four jumpers.

How the Grove ports map to pins

Port label	Pins it carries (yellow / white wire)	Notes
D2, D3, D4, D5, D6, D7, D8	Dn / Dn+1 (port D2 = D2 and D3)	two digital pins per port
A0, A1, A2, A3	An / An+1 (port A0 = A0 and A1)	analog in. Grove analog modules (light sensor, rotary sensor) use only the yellow pin, so neighbouring A ports stay usable
I2C (×4)	SCL / SDA (A5 / A4)	all four are the same bus, so pick any
UART	RX / TX (D0 / D1)	don't use: this is the USB serial link to the laptop

Five rules

1. **Set the VCC switch to 5V.** It's the small slide switch beside the RST button. The Grove LCD and the NeoPixel ring need 5 V. On an Uno, 5 V is also the logic level, so nothing needs a level shifter.
2. **A used port blocks its neighbour.** Port D2 uses D2 and D3, so port D3 must stay empty (its first pin D3 is taken). That's why both appliances use D2, D4 and D6. Exception: modules that use only the yellow pin (light sensor, rotary sensor) leave the neighbouring port free.
3. **Wire colours mean the same everywhere:** yellow = first pin, white = second pin, red = 5 V, black = GND.
4. **No pull-up resistors.** Buttons and switches connect a pin to GND, and the firmware turns on the Uno's internal pull-ups (`INPUT_PULLUP`). Pressed or closed reads LOW; released reads HIGH.
5. **MC-6: use the breadboard as a junction box**, outside the centrifuge beside the Uno (the Uno doesn't fit inside; wires run in through the open back). Feed its + and – rails from one Grove cable's red and black. **MW-1 needs no breadboard**: chain the button grounds and run one GND wire.



Grove cable colours. Drawing: made for this guide from the datasheet.

2 Parts

One card per part type. Pin names in **bold** are what you'll see printed on the part.



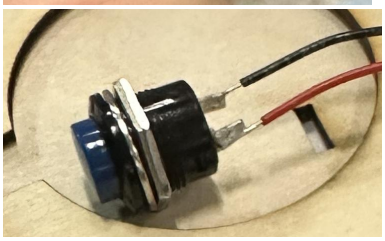
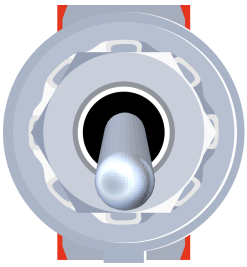


Photo: your part.

16 mm panel pushbutton (MC-6: START, STOP, TIME+; MW-1: all 5)

A momentary switch: the two lugs connect only while the cap is pressed. It mounts in a 16 mm hole with the nut behind the panel.

- **Pins:** two solder lugs, no polarity. Either lug can go to either side.
- **Connect:** one lug → the Arduino pin, the other lug → GND. Solder ~20 cm wires before mounting (as in your second photo), and cover each joint with heat-shrink.
- **Check:** with a multimeter on continuity, the lugs beep only while pressed.

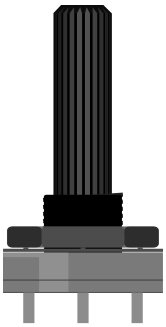


Drawing: Fritzing (CC-BY-SA 3.0).

SPDT toggle switch (MC-6: POWER)

A lever switch that stays where you put it.

- **Pins:** three lugs. The **middle** one is common; it connects to one outer lug or the other, depending on the lever.
- **Connect:** middle lug → GND, one outer lug → D5. Leave the other outer lug empty.
- **Check:** flip it and watch D5 change in the test sketch. If ON reads HIGH, use the other outer lug.





Drawing: Fritzing (CC-BY-SA 3.0). Photo: your part.

Speed potentiometer (MC-6: SPEED)

A 10 kΩ knob that gives 0–5 V on its middle pin, read on A0 as 0–1023.

- **Bare pot (WH148, as in the CAD):** three lugs. Middle (wiper) → A0 (yellow), one outer → 5 V (red), other outer → GND (black). If the dial reads backwards, swap the two outer lugs.
- **Grove Rotary Angle Sensor (second picture, your part):** plugs straight into port A0 with a Grove cable. It's a different shape from the WH148, so check it fits the control face first.



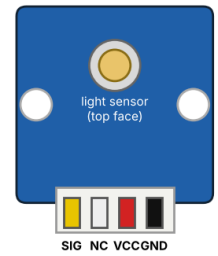
330 Ω: orange · orange · brown · gold

Drawing: made for this guide from the datasheet.

330 Ω resistor (optional)

Sits in the NeoPixel data line. It softens the signal edges and protects the first LED. The ring usually works without it; add it if colours flicker or the first LED acts up.

- **Pins:** two legs, no polarity. Bands: orange, orange, brown, gold. 220–470 Ω all work.



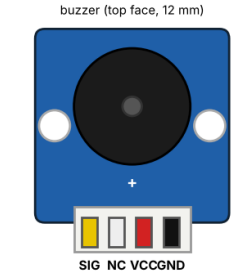
Grove Light Sensor v1.2
analog: brighter = higher reading
SIG → A port (yellow); NC unused

Drawing: made for this guide from the datasheet.

Grove Light Sensor v1.2 (MC-6: lid + tube, MW-1: door)

Measures how much light falls on it and gives an analog value, 0 (dark) to about 750 (bright room). It has no light source of its own. A lid, door or seated tube **shades** it, and the reading drops.

- **Pins:** Grove socket: SIG (yellow), NC (white, not connected), VCC (red), GND (black).
- **Connect:** a plain Grove cable to an **A** port. It can't go on a D port, because the value is analog.
- **Reading:** the firmware samples the open or empty level at power-up and treats a big drop (e.g. below 60% of that) as 'shut' or 'seated'. It uses hysteresis so the reading doesn't flicker at the edge, and it prints the raw number over serial for the verifier.
- **MC-6 tube sensor:** under well A, looking up through the tube. Light the NeoPixel ring a fixed colour while reading it, so the room light doesn't matter.
- **Mounting (now in the CAD):** MC-6 lid sensor on the inside of the left side panel, facing the lid fin; MC-6 tube sensor flat on the riser under well A, looking up. MW-1 door sensor standing on the sensor shelf, facing the door. Hot-glue or double-sided tape on the board's back.



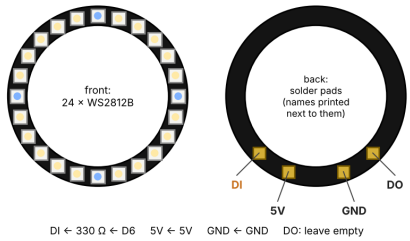
Grove Buzzer
SIG HIGH = sound (or tone() for pitch)
SIG → D port (yellow); NC unused

Drawing: made for this guide from the datasheet.

Grove Buzzer (MW-1)

A small buzzer with a driver transistor. SIG HIGH makes a sound; the firmware uses `tone()` on the same pin for a high key beep and a low 'refused' beep.

- **Pins:** Grove socket: SIG (yellow), NC (white, not connected), VCC (red), GND (black).
- **Connect:** a plain Grove cable to port **D8**. (Not an A port: it's an output.) Port D7 must stay empty, because its second pin is D8.
- **Sounds:** one short beep per button press, three beeps at DONE, two low beeps when START is refused.
- **Mounting:** on the base in the electronics bay, at the engraved BUZ mark, buzzer facing up. Hot-glue or tape.



Drawing: made for this guide from the datasheet.

NeoPixel ring, 24 × WS2812B (MC-6 only)

Glued under the clear rotor plate; shows ready (white), spinning (blue chase), done (green) and refused (red).

- **Pads (back): DI** data in, **5V**, **GND**, **DO** data out.
- **Connect:** D6 → DI (through the optional 330 Ω); 5V → + rail; GND → – rail. Leave DO empty. Solder 15 cm wires before gluing the ring in.
- **Brightness ≤ 50 of 255.** All 24 LEDs at full white draw about 1.4 A, which is more than USB gives the Uno. It browns out and resets.
- Optional: 470–1000 uF capacitor across 5V and GND at the ring.

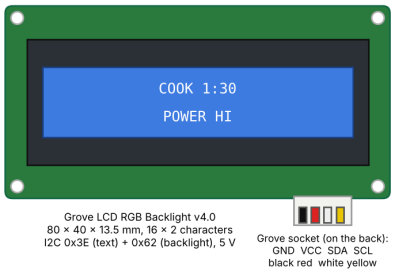


Drawing: made for this guide from the datasheet.

SSD1306 0.96 in OLED (MC-6 display)

128 × 64 monochrome display on the I2C bus, address 0x3C (some boards 0x3D).

- **Pins:** GND, VCC, SCL, SDA. Some boards swap GND and VCC, so wire by the printed names.
- **Connect with male-to-female jumpers** straight to the Uno header: GND → GND, VCC → 5V (power header), SCL → A5, SDA → A4 (analog header). Many Unos also have SCL/SDA sockets past AREF on the digital header, with the same signals.
- **Or** use a Grove-to-female cable in any I2C port: yellow SCL, white SDA, red VCC, black GND.

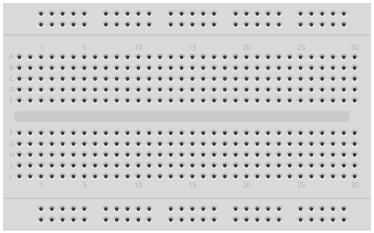


Drawing: made for this guide from the datasheet.

Grove LCD RGB Backlight v4.0 (MW-1 display)

16 × 2 character display with a colour backlight. Two I2C addresses: 0x3E (text) and 0x62 (backlight).

- **Connect:** a plain Grove cable from its socket to any I2C port. Nothing else to wire.
- Needs 5 V (VCC switch on 5V). Library: Seeed *Grove - LCD RGB Backlight* (`rgb_lcd.h`).



Drawing: Fritzing (CC-BY-SA 3.0).

Half-size breadboard (MC-6)

The MC-6 junction box, beside the Uno outside the centrifuge.

- The long rows marked + and – along each edge are the power rails. Feed them from one Grove cable's red (5 V) and black (GND).
- Each short column of 5 holes is joined. Plug a Grove cable's yellow or white male pin into a column, then the part's wire into the same column.

3 MC-6 centrifuge

Parts on this board: 3 16 mm panel buttons, the POWER toggle, the SPEED pot, the NeoPixel ring, 2 Grove Light Sensors (lid and tube), and the OLED. The Uno, shield and breadboard sit outside, beside or behind the centrifuge. Uses ports D2, D4, D6, A0, A1 and A2; the OLED goes to the header (or any I2C port). Port D8 and D6's white pin (D7) are free.

MC-6 centrifuge: Grove ports (top view, as in your photo)



* OLED: male-to-female jumpers to the Uno header (5V, GND, A4, A5), or a Grove-to-female cable in this port.

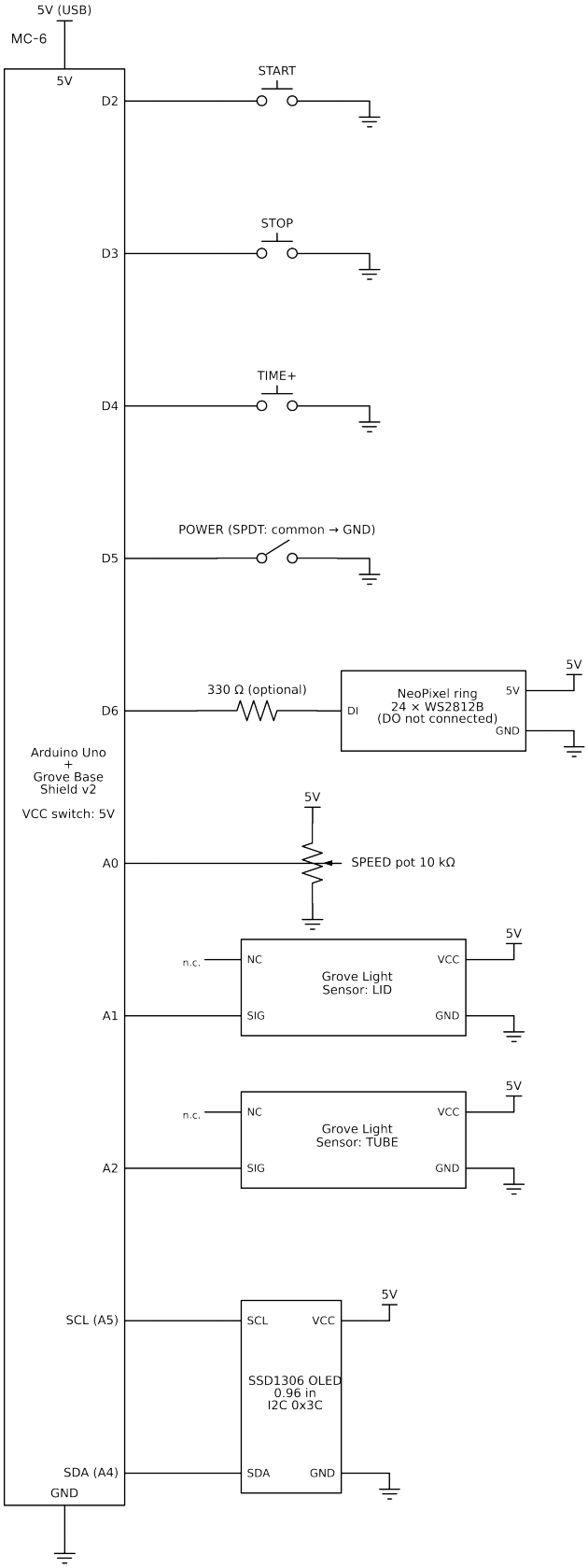
Part	Port · wire	Goes to
Button GND	D6 · black	the chained GND lugs of all 5 buttons
START button	D2 · yellow	one lug; other lug → – rail
STOP button	D2 · white (D3)	one lug; other lug → – rail
TIME+ button	D4 · yellow	one lug; other lug → – rail
POWER toggle	D4 · white (D5)	an outer lug; middle lug → – rail
NeoPixel ring	D6 · yellow	(330 Ω →) DI. Ring 5V → + rail, GND → – rail
SPEED pot	A0 · yellow / red / black	wiper / one end / other end. Tape the white pin (A1)
Lid light sensor	A1 · Grove cable	plugs straight in
Tube light sensor	A2 · Grove cable	plugs straight in
OLED	header: 5V, GND, A5, A4	VCC, GND, SCL, SDA

Wiring order

1. Set the shield's VCC switch to 5V. Unplug USB while wiring.
2. Solder leads: 3 wires on the ring (DI, 5V, GND), 2 on the toggle (middle + one outer) and 2 on each 16 mm button. Mount the buttons in the control face (nut behind). Run all leads out through the open back to the breadboard beside the Uno.
3. Plug a Grove-to-male cable into port D6. Put its red pin in the + rail and its black pin in the – rail. Tape off its white pin (D7, unused).
4. Wire the ring: D6 yellow → a column → (330 Ω →) ring DI. Ring 5V → + rail, ring GND → – rail.
5. Port D2: yellow → START lug, white → STOP lug. Their other lugs → – rail.
6. Port D4: yellow → TIME+ lug, white → toggle outer lug. TIME+ other lug and toggle middle lug → – rail.
7. Port A0: the pot. If it's a bare WH148 on a Grove-to-male cable, tape the white pin, because A1 is the lid sensor. Or plug the Grove rotary sensor straight in.
8. Ports A1 and A2: lid and tube light sensors, each on a plain Grove cable.
9. OLED: 4 male-to-female jumpers to 5V, GND, A5 (SCL), A4 (SDA).

10. Tape off every unused Grove red/black pin so it can't touch anything. Then plug in USB and run the test sketch.

MC-6 schematic



MC-6 test sketch

Install *Adafruit SSD1306*, *Adafruit GFX* and *Adafruit NeoPixel* from the Library Manager. Open the Serial Monitor at 115200. Buttons print 0 or 1. The light sensors print their raw value and their state (1 = shaded). Power up with the lid open and no tube in, so the sensors learn the bright level. Then press each button, flip the toggle, close the lid, drop the tube in and turn the dial. The ring stays dim white (it lights the tube sensor) and turns blue while START is held.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <Adafruit_NeoPixel.h>

Adafruit_SSD1306 oled(128, 64, &Wire, -1);
Adafruit_NeoPixel ring(24, 6, NEO_GRB + NEO_KHZ800);
const int   IN[]   = {2, 3, 4, 5};
const char* NAME[] = {"START", "STOP", "TIME+", "POWER"};
const int   LS[]   = {A1, A2};                // light sensors: lid, tube
const char* LSN[]  = {"LID", "TUBE"};
int bright[2]; bool shaded[2] = {false, false};

int readAvg(int pin) { long s = 0; for (int i = 0; i < 8; i++) s += analogRead(pin); return s / 8; }

void setup() {
  Serial.begin(115200);
  for (int p : IN) pinMode(p, INPUT_PULLUP);
  if (!oled.begin(SSD1306_SWITCHCAPVCC, 0x3C)) Serial.println("OLED not found (try 0x3D)");
  ring.begin(); ring.setBrightness(40);
  ring.fill(ring.Color(255, 255, 255)); ring.show(); delay(300);   // same light as when running
  for (int i = 0; i < 2; i++) bright[i] = readAvg(LS[i]);         // lid open, no tube
}

void loop() {
  oled.clearDisplay(); oled.setTextColor(SSD1306_WHITE); oled.setCursor(0, 0);
  for (int i = 0; i < 4; i++) {
    int v = digitalRead(IN[i]);                                    // 0 = pressed / ON
    Serial.print(NAME[i]); Serial.print('='); Serial.print(v); Serial.print(' ');
    oled.print(NAME[i]); oled.print(' '); oled.println(v);
  }
  for (int i = 0; i < 2; i++) {
    int raw = readAvg(LS[i]);
    if (!shaded[i] && raw < bright[i] * 0.55) shaded[i] = true;    // hysteresis: 55% / 70%
    if ( shaded[i] && raw > bright[i] * 0.70) shaded[i] = false;
    Serial.print(LSN[i]); Serial.print('='); Serial.print(shaded[i]);
    Serial.print(" raw="); Serial.print(raw); Serial.print(' ');
    oled.print(LSN[i]); oled.print(' '); oled.print(shaded[i]); oled.print(" ("); oled.print(raw); oled.println(")");
  }
  int s = analogRead(A0);
  Serial.print("SPEED="); Serial.println(s);
  oled.print("SPEED "); oled.println(s); oled.display();
  ring.fill(digitalRead(2) == LOW ? ring.Color(0, 0, 255) : ring.Color(255, 255, 255));
  ring.show();
  delay(100);
}
```

4 MW-1 microwave

Parts on this board: 5 panel pushbuttons, a Grove Light Sensor (door), a Grove Buzzer, and the Grove LCD. Uses ports D2, D4, D6, D8, A0 and one I2C port. No breadboard. The Uno and shield sit on the base in the electronics bay.

MW-1 microwave: Grove ports (top view, as in your photo)



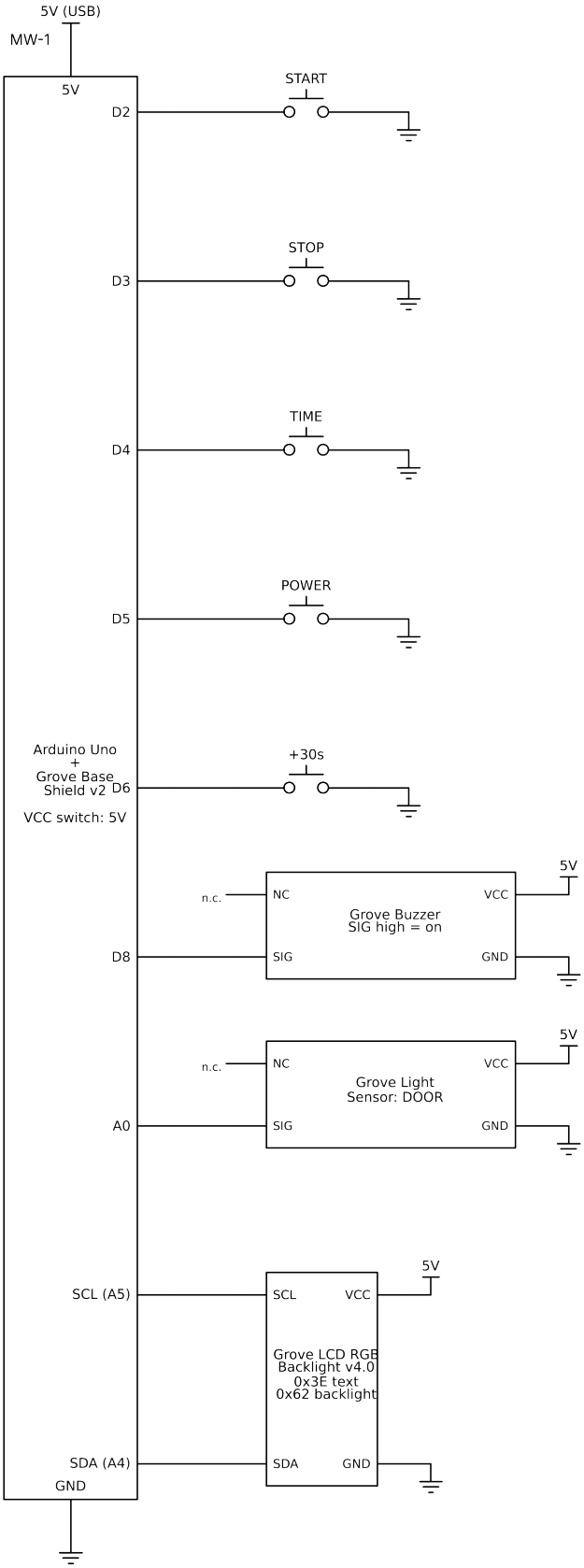
Part	Port · wire	Goes to
Grove LCD RGB v4.0	I2C · Grove cable	the LCD's Grove socket
Button GND	D6 · black	the chained GND lugs of all 5 buttons
START button	D2 · yellow	one lug; other lug → GND chain
STOP button	D2 · white (D3)	one lug; other lug → GND chain
TIME button	D4 · yellow	one lug; other lug → GND chain
POWER button	D4 · white (D5)	one lug; other lug → GND chain
+30s button	D6 · yellow	one lug; other lug → GND chain
Door light sensor	A0 · Grove cable	plugs straight in; mount it so the shut door shades it
Buzzer	D8 · Grove cable	plugs straight in; on the BUZ mark in the bay

Wiring order

1. Set the shield's VCC switch to 5V. Unplug USB while wiring.
2. Solder ~20 cm of wire to both lugs of each of the 5 buttons. Heat-shrink every joint. (If your button wires end in male pins, they can go straight into the header sockets D2–D6 instead of a Grove cable, with grounds to the – rail.)
3. Mount the buttons in the control panel (nut behind). Fix the light sensor where the shut door covers it, facing the door.
4. Port D6 Grove-to-male cable: black → the button GND chain. Tape its red pin (5 V) and white pin (D7, unused).
5. Port D2: yellow → START, white → STOP. Port D4: yellow → TIME, white → POWER. Port D6: yellow → +30s.
6. Chain all five buttons' other lugs together behind the panel with one wire, and join that to the D6 black pin.
7. Door light sensor: a plain Grove cable into port A0. Route it through the divider's 10 mm hole.
8. Buzzer: a plain Grove cable into port **D8**. Leave port D7 empty.

9. Plug the Grove LCD into any I2C port with a Grove cable.
10. Tape off the unused red/black pins of the D2 and D4 cables. Plug in USB and run the test sketch.

MW-1 schematic



MW-1 test sketch

Install *Grove - LCD RGB Backlight* (Seeed) from the Library Manager. Serial Monitor at 115200. Power up with the door **open** so the sensor learns the bright level. The top line shows the last button pressed. The bottom line shows DOOR SHUT (white backlight) or DOOR OPEN (red backlight) and the raw light value. Each button press also gives a short beep on the buzzer (D8).

```
#include <Wire.h>
#include "rgb_lcd.h"

rgb_lcd lcd;
const int   IN[]   = {2, 3, 4, 5, 6};
const char* NAME[] = {"START", "STOP", "TIME", "POWER", "+30s"};
const char* last = "-";
int bright; bool shut = false; int lastV[5] = {1, 1, 1, 1, 1};

int readAvg(int pin) { long s = 0; for (int i = 0; i < 8; i++) s += analogRead(pin); return s / 8; }

void setup() {
  Serial.begin(115200);
  for (int p : IN) pinMode(p, INPUT_PULLUP);
  lcd.begin(16, 2);
  pinMode(8, OUTPUT);                       // Grove Buzzer on port D8
  bright = readAvg(A0);                     // door open at power-up
}

void loop() {
  for (int i = 0; i < 5; i++) {
    int v = digitalRead(IN[i]);             // 0 = pressed
    Serial.print(NAME[i]); Serial.print('='); Serial.print(v); Serial.print(' ');
    if (v == LOW) last = NAME[i];
    if (v == LOW && lastV[i] == HIGH) tone(8, 2700, 60);   // beep on each new press
    lastV[i] = v;
  }
  int raw = readAvg(A0);
  if (!shut && raw < bright * 0.55) shut = true;      // hysteresis: 55% / 70%
  if ( shut && raw > bright * 0.70) shut = false;
  Serial.print("DOOR="); Serial.print(shut); Serial.print(" raw="); Serial.print(raw);
  Serial.println();
  lcd.setCursor(0, 0); lcd.print("Pressed: "); lcd.print(last); lcd.print("      ");
  lcd.setCursor(0, 1); lcd.print(shut ? "DOOR SHUT " : "DOOR OPEN "); lcd.print(raw); lcd.print("    ");
  if (shut) lcd.setRGB(255, 255, 255); else lcd.setRGB(255, 0, 0);
  delay(50);
}
```

5 Testing and troubleshooting

Symptom	Likely cause	Fix
A button always reads 0	A wire shorted to GND	Check for stray strands at the lugs
A button never changes	Wire on the wrong Grove pin, or the port next to a used one	Check the port map; yellow = first pin
Light sensor never changes state	Shading too weak, or it was calibrated while shaded	Power up with lid/door open and tube out; check raw values; adjust the 55% / 70% thresholds
Light sensor reads ~0 always	On a D port, or cable in the wrong A port	Use an A port; check the pin in the sketch
Tube sensor jumps with room light	Ambient light through the lid	Keep the ring lit at a fixed colour while reading
No beep	Buzzer not on port D8, or something in port D7	Use port D8; leave D7 empty. Try BUZZER_TONE = false in the firmware
OLED blank	Wrong address or SDA/SCL swapped	Run i2c_scanner; expect 0x3C (or 0x3D)
LCD lit but no text	VCC switch on 3V3	Set it to 5V
Ring dead	Wired to DO instead of DI	Move the data wire to DI
Board resets when the ring lights	Too much current over USB	Brightness ≤ 50, or add the capacitor
Upload fails	Something on D0/D1 (UART port)	Unplug the UART port

Power budget (USB, 500 mA)

Load	MC-6	MW-1
Arduino Uno	~50 mA	~50 mA
Display	OLED ~20 mA	Grove LCD ~60 mA
Grove Light Sensors	2 × ~0.5 mA	1 × ~0.5 mA
Grove Buzzer (while sounding)	—	~30 mA
NeoPixel ring, brightness 40–50	up to ~280 mA (all white)	—
Total	~350 mA	~140 mA

Only one board?

Not any more. The Uno has six analog inputs, and two of them (A4, A5) are the I2C bus. That leaves four, and the two appliances need four analog inputs: SPEED, lid, tube and door, plus the buzzer pin. It could just fit, but the two firmwares are separate, so use one board per appliance. (The I2C addresses wouldn't clash: OLED 0x3C, LCD 0x3E and 0x62.)

Credits

Arduino Uno, toggle switch, potentiometer and breadboard drawings: Fritzing parts library, CC-BY-SA 3.0 (fritzing.org). Photos of the shield, 16 mm button and Grove rotary sensor: your own. Other part drawings (OLED, Grove LCD, light sensor, buzzer, ring, resistor, cable) were made for this guide from the parts' datasheets.